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Morality in Our Mind and Across Cultures and Politics

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Abstract

Moral judgments differ across cultures and politics, but they share a common theme in our minds: perceptions of harm. Both cultural ethnographies on moral values and psychological research on moral cognition highlight this shared focus on harm. Perceptions of harm are constructed from universal cognitive elements—including intention, causation, and suffering—but depend on the cultural context, allowing many values to arise from a common moral mind. This review traces the concept of harm across philosophy, cultural anthropology, and psychology, then discusses how different values (e.g., purity) across various taxonomies are grounded in perceived harm. We then explore two theories connecting culture to cognition—modularity and constructionism—before outlining how pluralism across human moral judgment is explained by the constructed nature of perceived harm. We conclude by showing how different perceptions of harm help drive political disagreements and reveal how sharing stories of harm can help bridge moral divides.

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INTRODUCTION

Morality is fundamental to humanity. Questions of morality start wars, animate courtrooms, and lie at the center of everyday gossip. Which acts are right versus wrong seems obvious to people, but less obvious is the nature of moral judgment within our minds and across cultures and politics. Do all people fundamentally have the same moral sense? Or do different people hold importantly different values? The answer to both questions appears to be yes: Moral judgment is characterized by both differences and similarities, suggesting an apparent paradox.

Studies inspired by cultural anthropology find cross-cultural variation in the acts that people condemn. Tibetan Buddhists consider it immoral to gossip (Mumford 1989), while most Americans view it as a normal part of life. Most Europeans do not believe in witchcraft, but in Uganda, threatening another person by supernatural means is punishable by life in prison [Uganda Witchcraft Act 1957 (Chapter 124)]. In the United States, liberals and conservatives disagree about the immorality of abortion, illegal immigration, and countless other issues.

Despite these cross-cultural differences, studies inspired by cognitive science find striking similarities in moral cognition—the cognitive processes underlying moral judgments. Everyone's moral judgments are sensitive to intentionality, causation, and suffering (Barrett & Saxe 2021), explaining why, across cultures, intentional wrongs are seen as worse than accidental wrongs, directly causing harm is seen as worse than indirectly causing harm, and torture is seen as worse than shoving. Moral judgments are also sensitive to the mental capacities of those involved in an act (Gray et al. 2012), explaining why people everywhere forgive the bad behavior of toddlers more than adults. These universal moral considerations are present in the moral cognition of young children (Hamlin et al. 2008, Woo et al. 2017) and are embedded in moral codes from ancient Sumerian laws to the US Constitution, leading some to argue that all humans possess an innate and universal moral grammar (Mikhail 2007).

These two sets of findings—differences in cultural values and similarities in cognition—raise the question of how cultural diversity coexists with cognitive consistency. How can you get both moral relativism and moral universalism (Shweder 2012)? Resolving this seeming paradox requires appreciating a key principle of social psychology: Sitting between the collective behaviors of a culture and the implicit processes of cognition is our mind, the constellation of perceptions and subjective experiences that makes meaning of our social world and gives rise to different values, beliefs, and behaviors.

Definitions of social psychology often focus on how other people impact our thoughts, attitudes, and behaviors (Allport 1954), but social psychology is also defined as the study of subjective experience (Wegner & Gilbert 2000). According to this definition, to understand behavior and judgment, we need to understand what it feels like within our minds to behold our social world—how different acts appear, and how perceptions shape our judgments. This experience-centered approach to understanding the mind echoes classic ideas from sociology about how we subjectively construct our social reality (Berger & Luckmann 1966) and actively make meaning of acts in the world (Weber 1978).

Recognizing the power of subjective experiences helps to reconcile the apparent paradox between universal moral elements—intention, causation, and suffering—and cultural differences in moral condemnation. The cognitive machinery of morality is shared by all people, but cultures shape the interpretations of acts (Geertz 1973), determining how much they seem to involve intention, causation, and suffering. The more someone's cultural worldview leads them to see an act as harmful—involving intentionally caused suffering—the more they see it as morally wrong.

Different cultures construct different perceptions of what causes suffering (Shweder et al. 1997), allowing for moral diversity to arise from common concerns about harm. This is evident in the examples that we began with: Tibetan Buddhists, more than Americans, perceive gossip as causing sickness in communities, and Ugandans, more than Europeans, perceive witchcraft as a viable way to harm others. Even within the United States, people disagree widely about how much fetuses, the environment, or sacred books can truly suffer, and these different perceptions drive moral disagreements (Womick et al. 2024).

In this review, we unite the classic and modern studies that highlight the power of perceived harm in human morality. Our subjective experience of moral condemnation revolves around the perception of harm, and people become morally outraged when an act seems to cause harm. The ubiquity of harm in morality helps resolve the apparent paradox of universal cognition and cultural differences. Cultural pluralism in moral values arises from different psychological understandings of perceived harm, all of which are grounded in universal cognitive elements.

We first trace the history of harm in moral psychology, from studies on the moral development of children to the importance of harm in cross-cultural accounts of morality. Second, we explore the ubiquity of perceived harm across existing taxonomies of values, before resolving the apparent tension between universal concerns about harm and pluralist theories of moral values. Third, we explore modern work on dyadic morality that formalizes the role of perceived harm in moral cognition and helps make sense of patterns of moral judgments. Fourth, we conclude by showing how recognizing the universality of perceived harm across moral judgments helps to explain, and potentially bridge, moral disagreements across politics.

THE HISTORY OF HARM IN MORALITY

What makes an act good or bad? This question has troubled humans for at least as long as we have been writing books. Immanuel Kant [1956 (1788), p. 166] famously wrote that "two things fill the mind with ever and new increasing wonder and awe—the starry sky above me and the moral law within me," and a good place to begin an exploration of the moral mind is with moral philosophy.

Moral Philosophy

Philosophers use reason and logic to argue for normative rules of morality, exploring what makes an act objectively right or wrong. Although philosophers are interested in moral truths that exist independently from human minds, psychologists are quick to point out their moral reasoning is constrained by the structure of their moral minds (Cushman et al. 2012b), allowing us to glean insights about our moral psychology from moral philosophy.

Philosophers across diverse schools of thought have long recognized the centrality of harm to morality. Plato (*Republic* 335e) wrote that "in no case is it just to harm anyone," and his student Aristotle (*Nic. Eth.* 1136a.20) later wrote that "to act unjustly is simply to do harm voluntarily." Roughly two millennia later, utilitarian philosophers further formalized the importance of harm in morality, arguing that what makes an act good or bad is whether it leads to beneficial or harmful consequences (Bentham 1789, Mill 1861).

In 1789, Jeremy Bentham articulated the utility principle, arguing that to act morally is to act in ways that maximize pleasure and minimize pain. Bentham also argued that what determined someone's moral status—whether someone is deserving of rights and moral concern—was their vulnerability to harm. Discussing the moral status of nonhuman animals, he writes, "The question is not, Can they reason? nor, Can they talk? but, Can they suffer?" (Bentham 1789, p. 311).

John Stuart Mill (1861) elaborated on Bentham's ideas, expanding the notion of harm from obvious physical pain and pleasure to more psychological and social forms of harm. He argued that people could be harmed by acts that undermine their autonomy or opportunities, whether by extreme economic inequality or through societal structures that limit personal freedom [Mill 2004 (1848)]. In this way, Mill anticipated the broader understanding of suffering revealed in different cultures (Shweder et al. 1997) and emphasized by modern harm-based theories of moral judgment (Schein & Gray 2018). Mill also argued for the harm principle, a cornerstone of modern legal systems, liberal democracy, and libertarian politics: People should be free to do as they please so long as their actions do not harm anyone else [Mill 2002 (1859)].

What does the centrality of harm in moral philosophy tell us about our moral psychology? On one hand, the perspectives of philosophers can seem disconnected from the experiences of everyday people; after all, Kant once likened sex to sucking dry a lemon [Kant 1997 (1781)]. On the other hand, the moral intuitions of philosophers—the roots of their moral judgments—are often the same as those of everyday people (Gigerenzer 2008). Nietzsche argued that the biggest joke about Kant was that he spent his career invoking impossibly complex philosophical proofs to justify our most obvious moral intuitions: Do not lie, cheat, murder, or steal [Nietzsche 2018 (1886)]. Studies also find that the moral judgments of modern professional philosophers about classic moral dilemmas are influenced by morally irrelevant features such as order effects or the specific wording of the scenarios (Schwitzgebel & Cushman 2012). The ubiquity of harm across moral philosophy suggests a human moral psychology fixated on harm.

The moral judgments of philosophers are interesting to think about, but early moral psychologists sought insights about the moral mind from a group of people less focused on complex philosophies: children.

Classic Moral Psychology

Children can be selfish, egocentric, and mean, but they are also hardwired to be moral (Bloom 2013). Children care deeply about what is right and wrong, and by the time they become adults, they have acquired the rules of their society. The first moral psychologists were developmentalists who studied the moral judgments of children.

Developmental psychology initially focused on outlining the stages of moral development. Jean Piaget (1932) first argued that young children have a heteronomous morality, simply condemning acts that adults condemn, whereas older children and adults develop an autonomous morality, independently applying moral principles. Lawrence Kohlberg (1964) extended this work by outlining three levels of moral development, the highest of which involved making moral judgments by applying universal ethical or philosophical principles. For example, Kohlberg's participants were able to reason through complex moral dilemmas, like whether it was okay to steal medicine to save one's dying wife, by thinking through universal considerations like "who would be harmed?" and "what would happen if everyone did this?" Kohlberg (1964) argued that his subjects were better or worse at solving these dilemmas based on how far along they were in their moral development.

Kohlberg's one-time collaborator Carol Gilligan (1977) criticized his ideas, arguing that the proposition of universal moral principles—and the prospect of some people being better at morality—neglected the importance of cultural context and community and instead glamorized the morality of Western men free from family responsibilities. Gilligan argued for the ethics of care, emphasizing that moral judgments—especially those of women—focused on care, empathy, and relationships. The importance of harm to morality can be seen throughout much of this early developmental work, but subsequent research more directly highlighted its importance.

Working in the 1970s and 1980s, Elliot Turiel discovered that children made sense of what belonged in the moral domain—which acts were morally prohibited—by referencing harm. Turiel and colleagues found that children saw many acts as conventionally (or socially) wrong, like wearing pajamas to school (Nucci & Turiel 1978), but kids saw harmful actions, like punching a classmate, as morally wrong and worthy of condemnation and punishment (Tisak & Turiel 1984). Conventional violations like classroom pajama wearing were seen as allowable if permitted by an authority figure, but harmful acts like punching a classmate were seen as wrong no matter the circumstances (Tisak & Turiel 1988).

Turiel (1983) developed these discoveries into a general theory of moral judgment that emphasized the importance of welfare, rights, and justice. What made something morally wrong, and therefore worthy of condemnation or punishment, was whether it caused harm. According to Turiel and the testimony of his subjects, harm mattered most to morality when it was intentionally perpetrated by one person on another. Children agreed that slipping and falling in the rain might cause suffering, but it was not immoral like hitting a classmate (Tisak & Turiel 1984). Presaging what studies of moral cognition would later reveal (Abrams & Gray 2025), Turiel found that children were most likely to condemn acts involving intentional interpersonal victimization.

Despite the clear importance of harm to many moral judgments, Turiel's conclusions were soon challenged for two reasons. First, social psychologists critiqued the methods—the how—of classic developmental moral psychology, in which children were asked to reason through moral dilemmas, explaining their judgments. Social psychology would soon discover that formal explanations can sometimes be specious, with people often "telling more than they know," inventing inaccurate reasons behind their actions and judgments (Nisbett & Wilson 1977). The field would also embrace the affective revolution, learning that many judgments (Zajonc 1980)—including moral judgments (Haidt 2001)—are often rooted more in emotional gut reactions than in careful reasoning. These discoveries led some to question whether interviews focused on moral reasoning could reveal the nature of moral judgment (Haidt 2001).

Scholars were right to point out the limits of self-reports and the unreliability of introspection for understanding mental mechanisms. However, it remains valuable to ask people about their moral judgments. Recall that social psychology should explain our mental experience of the world (Wegner & Gilbert 2000), including our moral world. People are experts on how they perceive and make sense of morality, even if they can be naive to the cognitive processes that give rise to their perceptions. Self-reports remain the best way to make sense of people's rich perceptions and assumptions about different acts. Moreover, later work reveals that people's insights about the cognitive processes of moral judgment—especially as they connect to harm (Cushman et al. 2006)—are often correct.

The second challenge to Turiel's findings concerned the subjects—the who—of the work. Developmental psychologists of the time mostly interviewed Western children from affluent households, and contemporary psychologists acknowledge the limitations of drawing conclusions about human psychology from WEIRD (white, educated, industrialized, rich, democratic) people (Henrich et al. 2010). Cultural anthropologists argued that we should look further to understand morality (Shweder et al. 1987).

Cultural Anthropology of Morality

To look beyond the West, the anthropologist Richard Shweder traveled to the small temple town of Bhubaneswar, India in the 1980s, where he interviewed a sect of Brahmins about morality. Brahmins are a priestly caste in the Hindu varna system, and have cultural duties, roles, and spiritual traditions that Shweder hypothesized would give them a broader sense of morality than the Westerners interviewed by Turiel. Although Shweder recognized the limitations of Turiel's work, he also echoed it in key ways. First, his methodology was identical: He interviewed children and adults about their moral judgments and asked them to explain their reasoning. Second, Shweder was also interested in how his participants understood differences between moral and social (or conventional) violations.

Shweder interviewed Brahmin children and adults about the wrongness of different acts and compared these conversations to interviews with children in Hyde Park, a fairly affluent neighborhood in Chicago (Shweder et al. 1987). This methodology focused not on the mechanisms of moral cognition but on people's experiences of the moral world and the moral discourse they used to make sense of which acts were permissible versus forbidden.

Replicating Turiel's findings, Shweder found that both Americans and Brahmins condemned interpersonally harmful acts. Both Brahmins and Americans viewed kicking an innocent animal as immoral (Shweder et al. 1987), but Brahmins had moral concerns that seemed to extend beyond Americans' concerns about obvious interpersonal harm. They also condemned acts that undermined community bonds, established hierarchies, and duties to the group (Shweder et al. 1997). For example, Brahmins, but not Americans, considered it wrong for the husband to cook while the wife is home or for a widow to remarry (Shweder et al. 1987, 1997), because these acts were seen as threatening the fabric of the community.

Beyond concerns about interpersonal harm and community cohesion, Brahmin conversations about morality frequently touched on issues of purity, sanctity, and maintaining the natural order (Shweder et al. 1997). As the priestly caste, Brahmins were responsible for separating the sacred from the profane, and much of their lives were structured around maintaining the purity of their caste and their temple (Shweder et al. 1987). For example, when Brahmin schoolchildren returned home, they were required to change their clothes and shoes because these articles had become "polluted" during the day (Shweder & Miller 1985). This respect for purity also explained why Brahmins considered it morally wrong for an eldest son to eat chicken for the 10 days following his father's funeral—doing so prevented him from processing the father's "death pollution" (Shweder et al. 1987).

Looking across these interviews, Shweder and his colleagues (1997) suggested that discussions of morality centered on three important themes, or ethics: autonomy (concerns about direct harm, justice, and individual rights), community (concerns about hierarchy, duty, and the reputation of the group), and divinity (concerns about sacredness, purity, and pollution). These themes are best understood as narratives that people use to make sense of their moral world. They were not thought to reflect any specific cognitive mechanisms nor were they seen as deeply distinct, as explanations of immoral acts frequently touched on multiple ethics (Shweder et al. 1997).

There were several important conclusions drawn from Shweder's work. The first was that cross-cultural conversations about morality extended beyond the concerns about direct physical harm that seemed to dominate Western morality. Some scholars would go on to argue that the morality of Westerners is impoverished compared to other cultures, focusing narrowly on questions of individual rights and justice while neglecting questions of community and divinity (Rozin et al. 1999). But while the ethic of autonomy is clearly prioritized in the utilitarian philosophy of Mill and the participants of Turiel and Kohlberg, it is easy to find evidence of all three ethics in the West. Thomas Aquinas [1947 (1273)] was one of the most influential thinkers in Western thought, and he was also deeply religious and wrote about the moral importance of following God's natural law. Similarly, the Western philosophers Rousseau [1998 (1762)] and Hegel [1967 (1821)] both wrote about the importance of community bonds for living an ethical life.

Examining the judgments of modern Western secular atheists also reveals concerns for all three of Shweder's ethics. Progressives clearly value community, as they criticize those who neglect their homeowners' association or family responsibilities and express outrage at policies that seem to undermine their country. Likewise, left-leaning Westerners care about purity when it comes to preserving the sanctity of the natural world (Frimer et al. 2017) or purifying their body through juice cleanses and hot yoga.

Shweder's work revealed descriptive differences in the way that cultures described their moral judgments, but as we discuss, moral differences across cultures are often underlain by deeper similarities in cognition.

A second conclusion from Shweder's work was that if harm (i.e., autonomy) was just one of three major moral themes, then it could not fully explain morality. For example, the Brahmin prohibition of an eldest son eating chicken after a funeral seemed disconnected from the harm-based moral judgments of American children. Some interpreted this disconnect as revealing that considerations of harm were completely absent from many moral judgments (Haidt & Joseph 2004), but as we show, this interpretation is a mistake. A closer reading of both Turiel and Shweder reveals that these scholars considered harm to generally underlie moral concerns (Shweder 2012, Turiel et al. 1987). Each recognized that harm was in the eye (and culture) of the beholder. One person could see harm in ignoring community duties, while another person could see harm in actions that polluted the soul.

In the words of Turiel and colleagues, people from different cultures make different "informational assumptions" about which acts are harmful, and these assumptions shape moral judgments (Turiel et al. 1987, p. 189). For example, some cultures believe that each of us possesses an immortal soul vulnerable to harm, and these cultures are likely to moralize acts that defile these souls (Shweder et al. 1997). As we soon discuss, Shweder and colleagues' work on causal ontologies of suffering outlined different cultural understandings of harm and how they connected to morality (Shweder et al. 1997).

A third conclusion of Shweder's work was that a good way to make sense of morality—especially moral differences across cultures—is through taxonomies. In the decades since these ethnographies were published, scholars have proposed different lists of important moral values. Although those who propose these taxonomies often use them to argue against a common basis of moral judgments, we draw on the ideas of Turiel and Shweder to suggest that perceptions of harm underlie each of these moral taxonomies. Cultural differences in values can coexist with universal concerns about harm.

TAXONOMIES OF A HARM-BASED MIND

On Taxonomies

For as long as scholars have been studying the natural world, they have attempted to categorize, organize, and number it. Aristotle proposed that four fundamental elements made up all matter: earth, air, fire, and water. Physicists argue that there are four main forces (gravity, electromagnetism, weak nuclear forces, and strong nuclear forces), and chemistry charts out 118 chemical elements in the periodic table. These taxonomies allow scientists to make sense of the world by dividing entities according to important shared properties and differences between them. However, as we move up levels of analysis from studying physical matter to biological species to psychological judgments, the question of how to group and categorize becomes much less clear.

In 1735, Carl Linnaeus proposed a taxonomy that divided all living creatures into a hierarchy of different kingdoms, families, and species based on their appearance and behavior. As a religious man who drew on the Aristotelian idea of essences, Linnaeus argued that every species had a God-given essence that distinguished it from others, but later research revealed that these categories were less essential than commonly believed. Charles Darwin's theory of evolution by natural selection revealed a common process underlying all these taxonomized differences, and later work revealed that all creatures shared similar biological machinery. That descriptive differences are often underlain by deeper similarities is an important lesson for all taxonomies.

How many kinds of human minds are there? In his classification of species, Linnaeus also included four "varieties" of man, with essential racial and mental differences between them. Of course, scholars now acknowledge that racial categories are socially constructed and rightly find it odious to argue for mental taxonomies across race. Still, the historical appeal of this idea should make us cautious about claims that any psychological taxonomy truly carves nature at its joints.

Mental states are more contextual and subjective than physical forces or species, and so psychological taxonomies are better understood as ad hoc frameworks whose popularity waxes and wanes with the current theories of the day. Psychology has witnessed the rise and fall of many taxonomies, from Freud's conception of the id, ego, and superego [Freud 2010 (1923)] to Maslow's hierarchy of needs (Maslow 1943).

Even if all psychological taxonomies hinge on present-day cultural understandings, they provide useful language for reducing the complexity of psychological phenomena into a smaller number of units that can be studied. For example, the Big Five dimensions of personality provide a powerful system for describing individual differences (Costa & McCrae 1992), and the *Diagnostic and Statistical Manual of Mental Disorders* (Am. Psychiatr. Assoc. 2013) allows clinicians to diagnose mental illness and prescribe medication, even as psychologists acknowledge that the categories of mental disorders are not independent natural kinds (McNally et al. 2015).

Holding in mind both the utility and limitations of taxonomies, we turn to popular taxonomies of moral values.

Taxonomies of Morality and Moral Pluralism

Taxonomies of moral values embrace moral pluralism, the idea that people care about and distinguish between multiple values (Shweder & Haidt 1993). The existence of some moral pluralism is trivially true given that people discuss different values, captured with different words. Consider the mere existence of this list as proof of moral pluralism: honesty, integrity, loyalty, respect, punctuality, enthusiasm, innovation, accountability, optimism, modesty, sincerity, transparency, hospitality, dedication, inclusivity, harmony, resourcefulness, and benevolence.

Moral taxonomies grapple with the problem of how to take hundreds of different words used to capture different values and clump them together into a smaller number of categories—ideally between three and seven, our memory's upper limit for novel categories (Miller 1956). These categories are invented and labeled by researchers to make sense of when different value words are invoked across cultures. Because they grow out of explicit conversations about values, taxonomies do not reveal mechanisms of the mind but instead clumps of convenience. We review these taxonomies in order of the number of categories they propose—from three to four to five to six to seven—and then argue that each can be understood through the lens of perceived harm.

Taxonomy of three: The Big Three of morality (autonomy, community, divinity). Many of the taxonomies of modern moral psychology are inspired by the work of Shweder and his colleagues (1997), who proposed the Big Three themes of moral discourse that we reviewed above: autonomy, community, and divinity. These themes were created to make sense of moral differences between Americans and Brahmins and were argued to be fluid and overlapping. For example, discussions of immoral acts often touched on themes of both community and divinity, autonomy and community, or divinity and autonomy (Shweder et al. 1997).

Taxonomy of four: Relationship regulation theory. The cultural anthropologist Alan Fiske (2004) and his later collaborator Tage Rai (Rai & Fiske 2011) argued that there were four fundamental relational contexts that created four moral concerns: unity (caring for ingroups), hierarchy (respecting power differences), equality (reciprocity), and proportionality (punishment and reward are proportional to costs and benefits). Relationship regulation theory argues that moral judgments hinge upon these constantly changing relational contexts (Rai 2018, Rai & Fiske 2011). For example, people are expected to act differently when interacting with friends, a spouse, or strangers, and this gives rise to context-specific moral judgments. These different relational expectations can create moral conflict. For example, skipping work to take care of your sick child might pit your duty to your boss (hierarchy) against your duty to your family (unity).

Taxonomy of four: Intuitive ethics. Inspired by the Big Three, two of Richard Shweder's students, Jonathan Haidt and Craig Joseph, argued that there were four key ethics underlying morality: suffering, reciprocity, hierarchy, and purity. These scholars took a classic taxonomic approach to categorizing moral values by reading five major social science books and counting moral themes that were discussed by all five authors: suffering, reciprocity, and hierarchy (Haidt & Joseph 2004). They then added an additional theme, purity, based on Shweder's work on discussions of purity and pollution across cultures (Shweder et al. 1997). These four themes were argued to be innate moral concerns that evolved to help meet the challenges of group living. For example, concerns about reciprocity were argued to have helped groups by encouraging food sharing and discouraging selfishness, and concerns about hierarchy were viewed as rooted in primate systems for monitoring group status. In contrast to Shweder, who made no claims about the mechanisms of cognition, these intuitions were argued to be innately prepared psychological modules, paving the way for moral foundations theory (MFT).

Taxonomy of five: Moral foundations theory. Within three years, Haidt and Joseph's four intuitive ethics became five moral foundations: harm, fairness, ingroup, authority, and purity (Haidt & Joseph 2007). [Additional foundations have since been proposed, including liberty (Iyer et al. 2012), honor (Atari et al. 2020), and ownership (Atari & Haidt 2023)]. MFT emerged to help make sense of political and cross-cultural differences in morality, arguing that a liberal and Western bias in psychology had led researchers to mistakenly limit morality to harm and fairness (Haidt & Joseph 2007). MFT argues that American liberals primarily emphasize harm and fairness in moral judgments, whereas conservatives (and non-Westerners) have a more balanced morality of all five moral concerns, which gives conservatives a supposed advantage in morality (Haidt 2012). It further argues that each of these values is grounded in a distinct cognitive module, a little switch or taste bud in the mind (and perhaps even the brain) (Haidt 2012).

MFT is influential because it provides intuitive language to describe some moral differences across politics (Graham et al. 2009). However, the primary instrument for measuring moral foundations, the Moral Foundations Questionnaire (Graham et al. 2011), has been criticized for political bias in item wording (Gray 2025, Womick et al. 2024), narrowly operationalizing these broad constructs (Gray et al. 2022a) in ways that favor conservatives. For example, items assessing purity ask whether "chastity is an important and valuable virtue," and items assessing authority ask whether "men and women have different roles to play in society" (Graham et al. 2011). Conservatives are well known to care more about traditional gender roles (Horowitz et al. 2017) and sexual chastity (Hardy & Willoughby 2017), suggesting that these items tap into existing political differences rather than deep foundations that reveal a conservative advantage in moral values.

Research shows that liberals also care about the five foundations of morality when these questions are worded differently. For example, liberals care more than conservatives about obedience to progressive authorities [e.g., environmentalists and civil rights activists (Frimer et al. 2014)] and the purity of the environment (Frimer et al. 2017). Other studies show that when questions assessing moral foundations are stripped of politically biased content, liberals and conservatives endorse them roughly equally (Schein & Gray 2015). Yet other work—which we explore in the section titled Moral Cognition—shows that moral foundations are not deep cognitive foundations.

One popular application of MFT uses the Moral Foundations Dictionary (Graham et al. 2009), an ad hoc list of morality-related words, to count differences in moral language across contexts and on social media (Hoover et al. 2020, Kennedy et al. 2021, Sagi & Dehghani 2014). We suggest that—like Shweder's Big Three, which inspired it—this taxonomy does not tap into deep cognitive foundations but instead clusters of values and explicit moral rhetoric used by different communities.

Taxonomy of six: Model of moral motives. Janoff-Bulman and colleagues (2008) developed the model of moral motives (MMM) based on the idea that liberals are more approach-oriented (i.e., motivated to improve the status quo) and conservatives are more avoidance oriented (i.e., motivated to maintain the status quo). The MMM crosses this fundamental distinction between approach and avoidance with three social contexts: the intrapersonal (i.e., the self), the interpersonal (i.e., interactions with a specific other), and the group (i.e., society and the collective) (Janoff-Bulman & Carnes 2013). This two-by-three framework yields six cells of moral values including self-restraint (intrapersonal, avoidance), social justice (interpersonal, approach), and social order (group, avoidance) (Janoff-Bulman et al. 2008). This taxonomy is unique in its combinatorial approach to morality, crossing low-level motivational tendencies with higher-level social contexts.

Taxonomy of seven: Morality as cooperation. Morality helps humans to cooperate (Tomasello & Vaish 2013), and morality-as-cooperation theory (MAC) (Curry 2016) argues that moral values enable cooperation in groups and prevent societal collapse. MAC draws on game theory to identify seven types of cooperation, each proposed to give rise to a universal moral value including helping kin, being brave, and respecting property rights. Supporting this claim, studies find that these values are endorsed across many societies (Alfano et al. 2024, Curry et al. 2019).

Taxonomies of Harm and Ontologies of Suffering

Different taxonomies argue for a different number and variety of moral values, but many also argue for the underlying importance of harm. For example, both MAC and different iterations of MFT suggest that moral values arise to serve the functional purpose of helping individuals and societies (and by extension their genes) guard against harm.

These functionalist accounts are useful because they propose an ultimate explanation for which acts become morally condemned (because they are harmful) and also explain which values and virtues are seen as morally good (because they prevent harm). MAC argues that being a good cooperator prevents groups from failing, and a group that emphasizes cooperative values would have a clear advantage when competing with other groups (e.g., bravery helps win wars). MFT argues that moral values emerge to help solve challenges that could harm groups, like how to distribute resources without resorting to within-group violence—a challenge solved by following norms of fairness and respecting the decisions of authority figures. Likewise, concerns about purity help people avoid foodborne pathogens and sexually transmitted infections (Haidt & Joseph 2004).

Although psychologists agree that morality ultimately evolved to prevent harm (Tomasello & Vaish 2013), some question how much people within cultures psychologically connect different moral values to harm. Some taxonomies, like MFT, separate concerns about care and harm from other moral concerns, but a closer look at the cultural assumptions surrounding these values reveals the ubiquity of perceived harm. Drawing from the work of anthropologists, we suggest that various taxonomies of moral values might be better understood as taxonomies of perceived suffering.

Anthropologists recognize that cultural phenomena—including moral judgments—can be understood only by examining the webs of cultural significance that imbue them with meaning (Geertz 1973). Blowing out tiny fire sticks shoved into sugar paste seems bizarre until you recognize that birthday candles help to celebrate the passage of time.

People make meaning of many phenomena using folk theories, intuitive beliefs about how the world works (Gelman & Legare 2011). These theories help people construct working models of their world, from the physical laws of motion (McCloskey 1983) to why others think and behave the way they do (Wellman 1992). People also have folk theories about the nature of suffering, which connect deeply to their understanding of morality.

In Shweder and colleagues' cross-cultural interviews about morality, they documented how cultures had different causal ontologies of suffering—theories that made sense of how people were harmed (Shweder et al. 1997). For example, biomedical ontologies of modern Western medicine use germ theory to explain the suffering caused by diseases, while astrophysical ontologies explain suffering in life circumstances as fated through the alignment of the stars. Whether sickness is seen as stemming from salmonella or from Mercury in retrograde depends on the causal ontology.

These ontologies of suffering mirror Turiel and colleagues' different informational assumptions about harm (Turiel et al. 1987) and help to explain cultural differences in morality. Consider the violations of purity that Shweder highlighted, like the Brahmin prohibition against the eldest son eating chicken after his father's funeral. This act seems harmless to Westerners but not to Brahmins, who believe that a person's immaterial soul becomes polluted upon death and that their successful passage into the afterlife requires processing of this death pollution. It is the eldest son's duty to process his father's death pollution by avoiding "hot" foods like chicken for at least ten days (Shweder 2012). According to the Brahmins' causal ontology of suffering, shirking these rules by eating chicken harms the spirit of the father by trapping him in limbo.

The connection between cultural understandings of harm and moral judgments of purity violations was first discussed by Mary Douglas (1966) in her book *Purity and Danger*. Douglas traced the concepts of purity, pollution, and dirt across different cultures and found that notions of impurity were linked to perceptions of danger, a link that is echoed in recent work exploring moral judgments of purity (Fitouchi et al. 2022, Gray et al. 2022a).

Historically, even Westerners sometimes see danger in impurity, especially sexual impurity. Dr. John Harvey Kellogg (the inventor of Kellogg's cereal) saw harm in the "impure" act of masturbation, quoting an unnamed medical writer in his handbook on healthy living who claimed that "neither the plague, nor war, nor small-pox, nor similar diseases, have produced results so disastrous to humanity as the pernicious habit of Onanism" (quoted in Kellogg 1890, p. 233). Likewise, the hit singer–turned–gay-rights opponent Anita Bryant saw harm in the "impure" act of homosexuality, arguing that "[h]omosexuals cannot reproduce, they *must* recruit, *must* freshen their ranks. And who qualifies as a likely recruit: . . . a teenage boy or girl" (Bryant 1977, p. 203, emphasis in original). Although modern progressives see masturbation and being gay as harmless (and even positive), Kellogg and Bryant believed that these acts caused suffering to children and society and condemned them accordingly.

With the understanding that different individuals and cultures perceive harm differently, we can see the many taxonomies of different moral values in a new light. Violations of different clusters of moral values are each perceived—at least by those who endorse those values within a particular culture—as causing harm. These perceptions of harm are ultimately motivated by evolutionary or functional concerns, like protecting one's group or enforcing social order, but these harms are legitimately seen by people. Sometimes these perceived harms can be difficult for Westerners to understand, especially when cultures seem to approve of violence (Fiske & Rai 2014), like honor killings or duels. But anthropological analyses of these violent acts show that they are seen as guarding against broader harms like ostracism of one's family or the dissolution of community ties (Ashokkumar & Swann 2022). Even in the West, some amount of violence is morally permissible, like killing in war and self-defense, but only because these acts are perceived as preventing victimization.

The importance of understanding perceptions—of people's mental experiences of moral acts—is essential not only for making sense of cultural differences but also for understanding the machinery of the moral mind.

MORAL COGNITION

Across cultures, moral judgments often center on discussions of harm. However, the anthropological focus on conversation and explanation has been criticized for revealing little about our cognitive moral processes (Haidt 2001). To understand the cognition underlying moral judgments, we must explore two additional streams of research. First, we review classic cognitive psychology research on how category judgments are made, then apply these insights to moral psychology. Second, we review studies specifically exploring moral cognition.

Even as we move from the structured interviews of the cultural anthropologists to the tightly controlled laboratory studies of modern psychologists, we must be careful not to lose sight of the importance of our subjective moral minds—people's perceptions of the world. In fact, the conclusions of research on moral cognition mirror the conclusions of anthropological surveys: People condemn acts based on how much they are perceived to cause harm.

Before we explore the processes of moral cognition, we review two disagreeing perspectives—modularity and constructionism—about how moral pluralism arises from the moral mind. Understanding these broader ideas is crucial because the apparent paradox of cultural differences and cognitive similarities is only paradoxical within a modular view of the mind, where each different value is argued to come from a distinct mental mechanism. A constructionist account of the mind resolves this paradox by suggesting that cultural differences can arise by construction:

when a small set of common cognitive elements are combined and understood within different contexts.

Modularity: Distinct Experiences from Distinct Modules

Frameworks for the mind often coevolve with popular metaphors, and in the 1980s, many cognitive scientists understood the mind as a computer. These scientists and philosophers believed that the mind contained many discrete mechanisms or *modules* that performed specific functions in response to receiving specific inputs from the world (Fodor 1983).

A core assumption of cognitive modularity is isomorphism, a one-to-one mapping between a cultural-psychological phenomenon and a dedicated cognitive mechanism. For example, Paul Ekman (1992) argued for six basic emotions across cultures, each of which had a distinct cognitive mechanism (e.g., an anger circuit) with distinct triggers (e.g., blocked goals) and behavioral outcomes (e.g., aggression).

The idea of modularity was soon applied to morality, in part because emotions were increasingly seen as being important to moral judgments (Haidt 2001). Researchers began by transforming Shweder's Big Three themes of ethical discourse into three distinct cognitive mechanisms (Rozin et al. 1999). Eventually, the idea of distinct moral modules became MFT, which described foundations as "little switches in the brain" (Haidt 2012, p. 123) or distinct moral taste buds (Haidt & Joseph 2004). Concerns about harm, fairness, ingroup, authority, and purity were argued to be fundamentally distinct switches or tastes, both in people's discussions of morality and deep within their minds.

Now, 40 years after the rise of the computer metaphor of the mind, modern research questions the idea of cognitive modularity (e.g., Suhler & Churchland 2011). Emotions are not underlain by distinct neural circuits but instead by broad and interconnected functional networks (Lindquist et al. 2012), and there is no single distinct profile of any specific emotion in the mind or body. Not only are the physiologies of anger and fear similar (Lindquist et al. 2012), but the experience of anger is itself variable and can include both hot rage on the highway and cold seething in a work meeting.

Falsifying claims of modularity, studies find little distinctness between different foundations. Moral judgments of ingroup and authority scenarios correlate at $r = 0.88$, and purity and authority judgments correlate at $r = 0.80$ (Graham et al. 2011). These correlations are often higher than the correlations among scale items within the same foundations, suggesting that the supposed modules are not well-defined cognitive switches or tastes. If the taste of bitterness were correlated at 0.88 with the taste of sweetness, we would not consider them unique taste buds or experiences.

If discrete moral modules existed in the brain or mind, we might also expect more certainty about their number, but advocates of MFT argue that "there may be 74, or perhaps 122, or 27, or maybe only 5" (Graham et al. 2013, p. 58). Given that the number of proposed foundations has increased from three (Rozin et al. 1999) to four (Haidt & Joseph 2004) to five (Haidt & Joseph 2007) to six or more, it may be more useful to think about MFT as an ad hoc taxonomy of values than a cognitive theory that carves nature at its joints. Of course, these taxonomies can be useful for describing culture, like genres of music, but culture should not be conflated with cognition.

Reflecting this state of evidence, some recent formulations of MFT argue that foundations should be considered only "developmental constructs" (Graham et al. 2018, p. 215) or moral concerns that children are predisposed to worry about. This blurring of foundations allows for a greater role of harm in moral cognition, especially because, as MFT researchers propose, "concerns about Harm and Fairness are so widespread that they might be said to be universally used foundations of morality (upon which *cultures construct differing ideas as to what counts as harm*, or what kinds of distributions are fair)" (Graham et al. 2011, p. 15, emphasis added). The connection between harm and fairness is also obvious, as receiving unfair distributions—whether in terms of pay, freedoms, punishments, legal protections, or food distributions—causes people to suffer.

The idea that perceptions of harm are universal to morality and culturally constructed is consistent with another view of the mind, one that argues against modularity. This view is called psychological constructionism.

Constructionism: Combining Basic Ingredients in Context

Psychological constructionism embraces the complexity of psychological experiences and cultural pluralism but denies that psychological categories (e.g., different values) correspond to distinct modules or circuits in the brain. Instead, constructionist theories propose that mental experiences are constructed by broad domain-general brain networks that combine a basic set of cognitive ingredients within specific social contexts.

For example, the theory of constructed emotion (Barrett 2017) argues that different emotional experiences occur when our minds make sense of core affect (feelings of good or bad plus high or low activation) using conceptual knowledge of emotions and context. The same state of physiological activation could be experienced as butterflies in the stomach when you are about to propose marriage or as terror when fleeing a pack of wolves. Supporting the constructionist view of mental states, modern neuroscience finds that different emotions are not located in distinct brain regions (Lindquist et al. 2012, Vytal & Hamann 2010) but instead in overlapping brain networks that subserve many functions (Barrett & Satpute 2013, Lindquist & Barrett 2012).

Dovetailing with the work of cultural anthropologists, theories of psychological constructionism argue that context and culture help determine our experience of the world (Barrett et al. 2011, Brooks et al. 2016). For example, people in the collectivist culture of Japan experience the emotion amae, a feeling of deep dependence and trust, whereas the Hebrew feeling of galut—the pain of diaspora—stems from an intimate cultural connection to migration and exile. Constructionism provides for pluralism across cultures because there can be as many different mental experiences as there are different combinations of core ingredients and conceptual understandings.

Constructionist theories of the mind suggest that moral pluralism across individuals, contexts, and cultures coexists with universal cognitive ingredients in moral judgment. Consistent with this idea, researchers suggest that moral judgments are not located in distinct "morality regions" in the brain (Van Bavel et al. 2015) and instead involve many social-cognitive considerations including perceived intentionality (Cushman & Young 2011), the experience of affect (Nichols 2002), perceived suffering (Schein & Gray 2015), inferred causation (Mikhail 2007), and other social cognitive elements besides (Barrett & Saxe 2021, Gray et al. 2012).

A harm-based model of the moral mind is a constructionist theory (Cameron et al. 2015), because it suggests that harm is constructed from basic psychological ingredients (e.g., intention, causation, suffering) and is perceived based on cultural scripts and social understandings. Cultures have different ideas about which entities are intentional moral agents (Gray & Wegner 2010), who is vulnerable to harm (Womick et al. 2024), and which acts cause suffering (Shweder et al. 1997), and this allows for pluralism in the perception of harm—and immorality.

Understanding the constructed nature of the moral mind helps provide "universality without the uniformity" (Shweder 2012, p. 5)—cultural differences from cognitive universals—and is also consistent with classic cognitive research on how the mind makes judgments more broadly.

How the Mind Makes Judgments

Moral judgments often seem like a special kind of judgment, accompanied by visceral feelings of outrage (Hutcherson & Gross 2011) and driving powerful phenomena like punishment (Shultz et al. 1986), ostracism (Williams 2007), and social movements (Sevelsted & Toubøl 2023). However, research finds that—cognitively—these judgments are best understood through the lens of general categorization judgments (McHugh et al. 2022). When people are judging the immorality of an act, they are deciding how well an act belongs to a category—the category of immorality.

Recognizing moral judgment as a process of categorization allows us to apply decades of cognitive psychology to understand moral cognition. This research finds that categorization judgments occur through the process of template matching (Rosch 1978). When we judge whether something is a member of a category—like "is this furry object a dog?"—we compare it to how well it intuitively matches our mental template of the category. This template is an internal representation of the category—also called a schema or a prototype—that includes the key features that denote category membership. The cognitive template of a dog might consist of a friendly, furry mammal with a hanging tongue and an expressive face.

The closer something in the world seems to match our internal template of the category (being a prototypical category member), the stronger—and quicker (Rosch & Lloyd 1978)—our judgment that it belongs to that category. This is why a golden retriever seems more doglike than a hairless Xoloitzcuintle. Similarly, when someone asks, "Does my partner belong to the category 'good spouse?'" they compare their partner to their mental template of a good spouse, perhaps someone kind and loving like their parents. When a doctor wonders, "Does this lump on a radiology scan belong to the category 'cancerous tumors?'" they compare it to their mental template of cancerous tumors.

Some suggest that categorization is the fundamental basis of all cognition: Surviving in the world requires we that we construct mental concepts of "good foods to eat," "things to run away from," and "people to befriend" (Harnad 2005), and templates represent these concepts. In fact, templates represent *all* our concepts, from snakes to airplanes, and are also used to categorize other people, like whether someone is attractive, intelligent, or belongs to a specific race. Again, the closer someone matches a template or prototype, the more powerfully and quickly we categorize them as part of that category or concept.

Importantly, research reveals that these cognitive templates are fuzzy representations of categories, meaning that things in the world can be better or worse fits for them (Murphy 2004). This fuzziness explains why some examples match a template only somewhat, like penguins as birds. We agree that a penguin is technically a bird, but it is not a very good bird. The fuzzy nature of mental templates also explains why we often just know it when we see it, and it can be difficult to articulate the rules that govern whether something fits a category. Even without being able to formally articulate what makes something a sport, rugby seems closer to the prototype of sport than chess or hopscotch.

People universally rely upon cognitive templates when making judgments about the world, but these templates also hinge on an individual's experiences and their cultural surroundings. For example, people in different cultures disagree about whether insects belong to the category of food and whether heavy metal falls into the category of good music. Cognitive templates are both adaptable and context dependent, just like moral judgments.

Given that all categorization judgments rely on template matching, we can ask, What is the nature of our template of immorality?

A Harm-Based Moral Template

Modern research on moral cognition suggests that our mental template of morality centers on harm. Harm can mean many things, but the paradigmatic immoral harm is interpersonal harm that violates the norms of society. More colloquially, people seem to morally condemn victimization—when one entity (a person or group) intentionally causes another to suffer.

The theory of dyadic morality (TDM) (Schein & Gray 2018) formalizes the psychological structure of harm/victimization, arguing that moral judgments occur by comparing acts to a cognitive template consisting of a dyad of an intentional agent (iA) causing damage ($^{d}\rightarrow$) to a vulnerable patient (vP), or $iA^{d}\rightarrow vP$. TDM argues that the more an act seems to intuitively match this cognitive template of harm, the more strongly it is judged as immoral. Put simply, the more an act seems to victimize someone vulnerable, the more immoral it seems. A continuum of perceived moral wrongness is driven by a continuum of perceived harm.

These three cognitive elements (an intentional agent, a vulnerable patient, and a causal link between them of suffering) are combined by the mind to form a gestalt representation of victimization, which is bolstered by the presence of each independent element. The more an act seems to involve each of intentionality, causation, and suffering, the more immoral it seems. For example, studies reveal that the more a wrongdoer seems like an intentional agent capable of thinking and planning, the more robustly they are condemned (Schein & Gray 2015), and this is reflected in the legal prerequisite for guilt: *mens rea*. The more someone seems like a vulnerable victim capable of suffering, the more people condemn damaging acts against them (Womick et al. 2024), which is why it seems worse to kick a puppy than a professional wrestler. The more an act directly causes damage, the more immoral it seems (Cushman et al. 2006), explaining why pushing someone off a cliff is worse than failing to maintain a fence at the top of a cliff, which causes someone to fall off a cliff.

At the center of our fuzzy moral template of victimization sit the most prototypical immoral acts, like intentionally kicking a newborn puppy. Acts like this are uncontroversially immoral because everyone agrees that they intentionally cause a vulnerable entity to suffer. As acts move from the clear center of the template to the fuzzy boundaries, there is more room for disagreement, as people see different amounts of intention, causation, and suffering in acts. Did your mother-in-law intend that backhanded insult at dinner? Can controversial campus speech cause trauma? Fuzziness around perceived harm creates space for moral disagreement.

Direct empirical support for a harm-based moral template comes from modern cognitive psychology studies. These studies attempt to decipher the structure and mechanisms of moral cognition by analyzing people's patterns of judgment. These methods use tightly controlled experiments with precise measurements to study people's automatic moral intuitions about different acts, and we explore several converging streams of evidence below.

Harm is most cognitively accessible. The simplest way to discern the structure of our moral template is to ask people to list examples of immoral acts. For all categories, items that most closely resemble the category prototype are the most cognitively accessible, which is why people are more likely to list cow as an example of a mammal than platypus (Rosch & Lloyd 1978). In a study that asked participants to list the first immoral acts that came to mind, participants most frequently listed interpersonally harmful acts like murder and assault, suggesting a moral template of dyadic harm (Schein & Gray 2015).

Harm is rapidly perceived. Moral judgments occur rapidly and intuitively (Haidt 2001), and a second way of studying the cognitive structure of morality is to put participants under time pressure or measure their reaction times as they make moral judgments. Concepts are more quickly categorized when they more closely match a category template (Rosch 1978), which is why people quickly identify a turkey club as a sandwich but take longer to decide if a hot dog is also a sandwich.

If moral judgments are intuitive and rely on a template of perceived harm, then harm should also be rapidly perceived—and it is. Although considerations of harm were once thought to be objective and reasoned, modern neuroimaging reveals that perceptions of harm are processed by

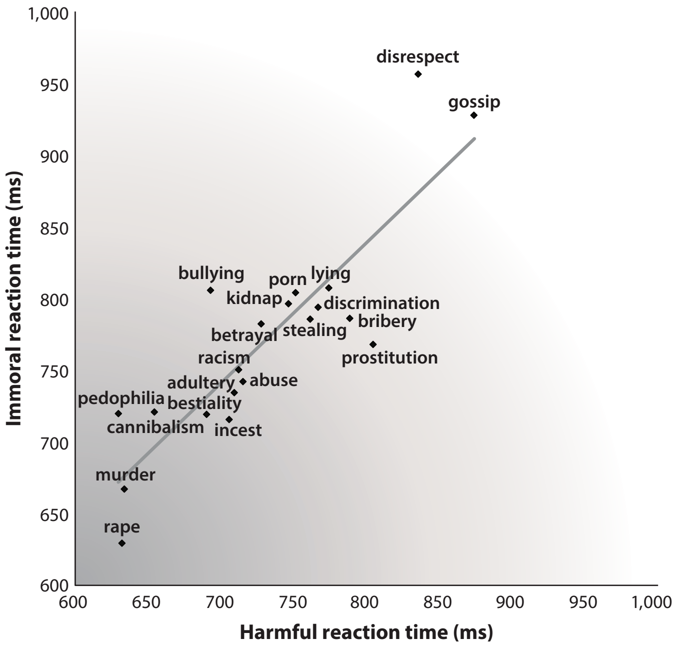

Figure 1

Evidence for a harm-based moral template. The speed at which participants judge an act as harmful tracks closely with the speed at which they classify it as immoral. Figure adapted from Schein & Gray (2015).

the brain within milliseconds (Decety & Cacioppo 2012). Studies also reveal that people quickly judge acts as harmful, and the speed at which people judge an act as harmful tracks closely with the speed at which they judge an act as immoral (Schein & Gray 2015). This alignment is exactly what a harm-based template predicts: The more accessible (i.e., obvious) an act's harmfulness, the more quickly it is judged as immoral (**Figure 1**).

Another study formally tested the idea of moral judgment as harm-based template matching (Ochoa 2022). Participants rated the immorality of various scenarios drawn from all five moral foundations. The scenario features that best predicted moral condemnation were "a potent and negative event directed at a very weak object," in other words, intentional dyadic harm (Ochoa 2022, p. 6). This study then had participants classify (under time pressure) whether or not a scenario was harmful, and whether or not it was immoral. Across 9,200 trials and 25 different moral scenarios, they found that the more closely a scenario resembled a prototype of dyadic harm (e.g., a person kills a child) the quicker people were to rate it as both harmful and immoral.

Harm is ubiquitous across moral judgments. These previous studies looked at dichotomous judgments (e.g., harmful versus not harmful), but harm and morality are perceived on a continuum: Genocide (very harmful) seems worse than stealing (somewhat harmful), which seems worse than surprise-tickling a friend (not harmful). If moral judgment relies on a harm-based template, the continuum of moral condemnation (from less to more immoral) should be almost perfectly predicted by a continuum of perceived harm (from less to more harmful).

In a direct test of this hypothesis, researchers had participants rate the immorality and harmfulness of various different acts taken from scenarios specifically designed to tap into different moral foundations. Supporting the predictions of a harm-based moral mind—and arguing

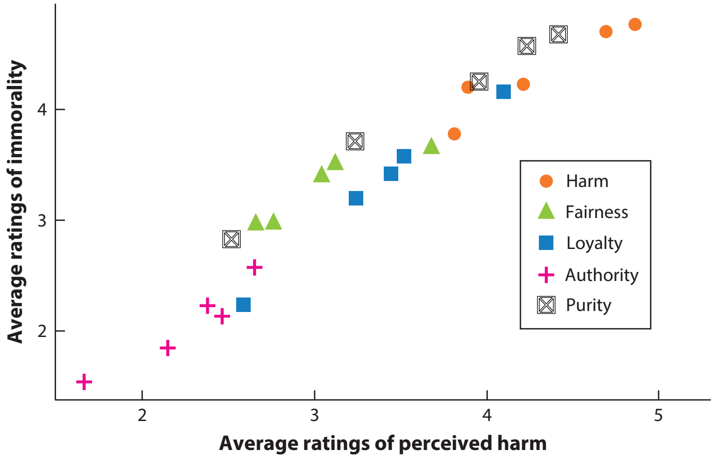

Figure 2

Additional evidence for a harm-based template. Ratings of immorality are almost perfectly predicted by ratings of harm, no matter which moral foundation the act comes from. Data and original figure from Ochoa (2022); figure adapted from Gray et al. (2022b).

against claims of distinct cognitive modules—moral condemnation across all acts was almost perfectly predicted by how much they were perceived to cause harm or victimization, regardless of whether they touched on values of care, fairness, loyalty, authority, or purity (Ochoa 2022) (**Figure 2**).

Studies of moral cognition often involve scenarios with anonymous strangers (Hester & Gray 2020) or strange events like sex with frozen chickens (Gray & Keeney 2015). These scenarios have some benefits, like providing experimental control or homing in on specific moral content, but it is fair to question how well a harm-based template makes sense of moral judgments in the real world (Yudkin et al. 2023), where considerations of relationships and identity loom large.

Although more naturalistic research is always welcome, one study analyzed the presence of moral condemnation in a massive set of 23,022 articles in Vox (left leaning) and 23,781 articles in Breitbart (right leaning) spanning many different topics and ideologies (Ochoa 2022). Analyses revealed that the likelihood of a sentence containing negative moral words, as assessed by the Moral Foundations Dictionary (Graham et al. 2009), was predicted by how much the sentence described dyadic harm. Across diverse acts in the lab and in everyday life, perceptions of harm best make sense of moral judgments, no matter what variety of value an immoral act seems to violate.

Harmless wrongs do not exist. That moral judgments are predicted by perceptions of harm allows us to make sense of the moral condemnation of harmless wrongs, acts that seem to lack a suffering victim but are nonetheless condemned. The most famous harmless wrong scenario describes consensual sex between two siblings named Mark and Julie. Many people condemn this scenario even though it was carefully written to be harmless, by specifying that Mark and Julie used contraceptives, kept it a secret, and had a great time (Haidt et al. 2000).

Although the Mark and Julie scenario is argued to be objectively harmless (Haidt et al. 2000), our tour through anthropology and its ontologies of suffering suggests that harm is better understood as a subjective perception (Schein & Gray 2018). Moreover, research on harm-based cognition finds that harm is rapidly and automatically perceived along a continuum, allowing it to persist to differing degrees even when others consider it objectively absent. Just as people can viscerally feel afraid of flying even if plane travel is quite safe, people can viscerally feel that a harmless action still involves some amount of harm. In one study, participants who discharged a fake gun at an experimenter's head experienced a visceral aversion even when the researcher told them it was harmless (Cushman et al. 2012a), and visceral perceptions of harm were shown to drive moral judgments.

In the two decades following the original formulation of harmless wrongs, much research has shown that people actually do perceive these acts as harmful. Studies using the supposedly objectively harmless Mark and Julie scenario find that even when researchers explicitly tell participants that the act is harmless, participants continue to believe that it could cause some harm, and this perception predicts their moral condemnation (DeScioli et al. 2012, Royzman et al. 2015, Stanley et al. 2019).

Moral foundations theorists argue that discussions of harm in harmless wrongs are mere after-the-fact rationalizations of intuitive non-harm-based judgments (Haidt et al. 1993), but another study finds that people perceive more harm in consensual incest when they are put under time pressure, showing that perceptions of harm are automatic and intuitive rather than post hoc rationalizations (Gray et al. 2014). Other studies find that people perceive victims in harmless acts like suicide, cannibalism, drug use, grave desecration, homosexuality, abortion, flag burning, and eating dog meat (Descioli et al. 2012). Harmless wrongs do not seem harmless to those who condemn them.

Advocates of harmless wrongs in moral judgment sometimes argue that negative gut feelings, including disgust, can drive moral condemnation of ostensibly harmless acts (Haidt et al. 2000). Two points bear mentioning. First, harm is itself affective. People are hardwired to become upset, uncomfortable, and angry when witnessing the victimization of someone vulnerable (Blair 1995), and so feelings of negative affect are intimately connected to perceptions of harm.

Second, studies show that feeling negative affect does not inherently drive moral judgment. Causally inducing people to feel negative affect or disgust does not reliably harshen moral condemnation (Landy & Goodwin 2015). There are many disgusting events in the world, but what distinguishes the disgusting and moral (e.g., changing a gross diaper) from the disgusting and immoral (e.g., cannibalism) is the perception of harm (Schein et al. 2016). One pharmacological study explicitly evaluated the relative impact of affect and harm on moral judgments by administering the affect-reducing drug propranolol and found that harm, but not negative gut feelings, directly drives moral condemnation (Gray et al. 2022b). Negative feelings connect to moral judgment by providing information [e.g., affect-as-information theory (Schwarz & Clore 2003)] about the harmfulness of ambiguous, strange, or disgusting acts (Gray et al. 2022b).

Studies of moral cognition find that all moral judgments involve a harm-based template, but moral disagreement clearly exists. How can a shared harm-based moral mind help us understand—and resolve—political divisions?

UNDERSTANDING AND BRIDGING MORAL AND POLITICAL DIFFERENCES

Political polarization is at an all-time high in the United States (Iyengar et al. 2012) and in many countries across the world (Hobolt et al. 2021). Many of these disagreements are moral disagreements. People see the other side as stupid (Hartman et al. 2022b) and as lacking a basic moral compass (Puryear et al. 2022). This moral conflict has profound consequences for society, driving distrust in government (Sasaki 2019), furnishing support for antidemocratic practices (Braley et al. 2023, Pasek et al. 2022), and destroying community relationships (Dunn 2020).

Fortunately, understanding the shared importance of harm to moral judgments—in cognition and across culture—can help us make sense of why these conflicts exist and how we can resolve them.

A harm-based moral mind suggests that moral and political disagreements are really disagreements about harm. Everyone cares about protecting victims from harm, but as Turiel and Shweder pointed out decades ago, different people make different assumptions about harm (Shweder et al. 1997, Turiel et al. 1987). New research provides empirical evidence for this idea, showing that moral disagreement across politics is in part grounded in different assumptions of vulnerability (AoVs) (Womick et al. 2024).

Assumptions of Vulnerability Explain Political Disagreement

AoVs are perceptions about who or what is especially vulnerable to harm, mistreatment, and victimization (Womick et al. 2024). They are assumptions because we lack direct access to the experiences and mental states of others, leaving us to infer who is most capable of suffering (Dretske 1973, Leudar & Costall 2004). We may all agree that babies are more vulnerable to suffering than Navy Seals, but between these two extremes, ambiguity exists. A recent set of studies by Womick and colleagues (2024) measured AoVs using three face-valid questions about a target, *X*:

1. *X* is especially vulnerable to being harmed.
2. *X* is especially vulnerable to mistreatment.
3. *X* is especially vulnerable to victimization.

Responses to this measure reveal that American liberals and conservatives have similar AoVs about the obviously vulnerable, like puppies, and the obviously less vulnerable, like professional wrestlers, but they disagree about the relative vulnerability of targets related to controversial political topics. Liberals are more likely than conservatives to view undocumented immigrants, gay people, and the environment as vulnerable to harm. Conversely, conservatives are more likely than liberals to view state troopers, the American flag, and the Bible as vulnerable. These different assumptions help explain differences in moral judgments, like why conservatives get more outraged than liberals at burning the American flag.

Other work reveals a causal connection between AoVs and moral judgment. For example, people condemn acts than harm more vulnerable victims (e.g., children and those with disabilities) compared with less vulnerable victims [e.g., adults (Gray & Wegner 2009)], and experimentally manipulating people's assumptions about the vulnerability of ambiguously vulnerable targets (e.g., a female CEO) impacts moral judgments toward them (Womick et al. 2024).

At least in the US, many political debates can be understood via AoVs of four ad hoc clusters: the Environment, the Othered, the Powerful, and the Divine. The Environment (e.g., coral reefs, rainforests, planet Earth) focuses on the vulnerability of the natural world. The Othered (e.g., transgender people, undocumented immigrants, Muslims) refers to people who are outside the dominant power structure, often called marginalized or disadvantaged people. The Powerful (e.g., authority figures, corporate leaders, state troopers) are people in positions of status or authority. The Divine (e.g., God, Jesus, the Bible) refers to sacred entities and holy books, which are often described as "alive and active" (Hebrews 4:12).

Across studies, most Americans—regardless of their political ideology—agreed on the order of vulnerability of these targets, rating the Othered and the Environment as most vulnerable, followed by the Powerful and the Divine. However, these studies also revealed important political differences in perceptions of vulnerability. Compared with conservatives, liberals tended to view the Othered and the Environment as more vulnerable to harm, perhaps explaining why liberals

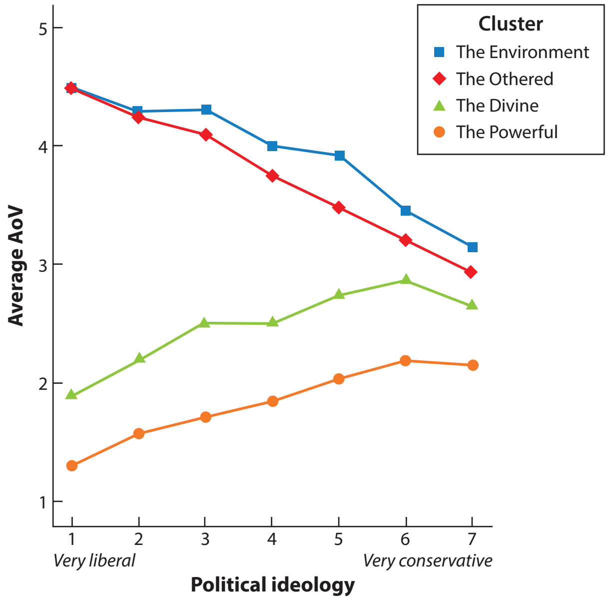

Figure 3

Assumptions of vulnerability (AoVs) for four clusters of targets across two different samples. These AoVs predict moral judgment and political disagreement. Figure adapted from Womick et al. (2024).

are more likely to support environmental protections [e.g., bans on offshore drilling (Mukherjee & Rahman 2016)] and social justice causes that seek to help marginalized groups [e.g., affirmative action (Jones 2005)]. Compared with conservatives, liberals also rated the Powerful and the Divine as less vulnerable to harm, perhaps explaining why liberals are more willing to endorse the slogan "eat the rich" and get less upset when someone defaces a Bible. The overall trends in AoVs across politics are displayed in **Figure 3**.

Zooming out from these specific differences reveals an overall trend distinguishing between the AoVs of liberals versus conservatives. Liberals are more likely to amplify differences in vulnerability, elevating the vulnerability of the typically vulnerable (e.g., marginalized groups) and reducing the vulnerability of the typically less vulnerable (e.g., corporate leaders). This amplification explains the progressive narrative that there are two broad groups in the world: invulnerable oppressors and the vulnerable oppressed (Freire 2005).

Conservatives are more likely to diminish differences in vulnerability, believing that all people are relatively equal in their susceptibility to suffering. This idea aligns with the conservative ideals of individualism and personal agency (Fine 1992), which suggest that all people are born equal in their ability to prosper, work, and suffer [US Declaration of Independence (1776)].

These left–right differences in AoVs help explain different social movements across politics. Liberals are more likely to support Black Lives Matter, perhaps because they see Black people as much more vulnerable to victimization than White people. Conversely, conservatives are more likely to support All Lives Matter, perhaps because they see all people as roughly equally vulnerable to victimization.

Acknowledging different perceptions of harm across politics is key for understanding how descriptive moral differences arise from a shared harm-based moral mind; it is also key for understanding how to bridge moral and political divides.

Bridging Divides by Sharing Stories of Harm

There are many strategies for bridging moral divides (Hartman et al. 2022a), but the average American believes that the best way to find common ground across political differences is to emphasize facts and statistics (Kubin et al. 2021). Unfortunately, this intuition is wrong: Launching facts at the other side often fails to foster respect and sometimes backfires (Nyhan & Reifler 2010). Instead, new research reveals that the best way to bridge moral divides is to connect with the other side through personal stories of harm.

Humans evolved telling stories (Gottschall 2013), and research shows that personal stories of harm create respect across political disagreements (Kubin et al. 2021, 2023). In one study, participants read about a hypothetical person who disagreed with them on an important political issue (e.g., taxes, gun control) either because of facts they read (e.g., statistics showing that gun control laws reduce gun deaths) or because of a harmful personal experience (e.g., their loved one was killed in a mass shooting) (Kubin et al. 2021). When opponents grounded their moral stance in a personal story of harm, as opposed to statistics, participants viewed them as more rational, had more respect for their stance, and were more willing to interact with them.

The power of stories of harm to bridge divides holds across contexts and platforms. In another study, research assistants had face-to-face debates about political issues with passersby on a college campus (Kubin et al. 2021). Political opponents were treated with more respect (behaviorally coded from audio recordings) when they grounded their stance in a personal story of harm versus facts. Similarly, an analysis of 137 interview transcripts found that guests on Fox News and CNN were treated as more rational by the host when they spoke more about personal stories of harm. In addition, an analysis of 300,978 comments on 194 YouTube abortion opinion videos found that comments were more positive when the videos emphasized personal experiences of harm, rather than facts.

Of course, facts are important for rational discourse, and other research shows that statistics can be paired with personal experiences to reduce political animosity and dehumanization of the outgroup (Kubin et al. 2023). Sharing personal experiences also reduces the desire to censor opposing moral opinions, because people want to censor things that seem harmful and false (Kubin et al. 2024), and personal stories seem truer than statistics (Kubin et al. 2021).

Though moral disagreements often seem intractable, there are many potential interventions that can bridge divides (Hartman et al. 2022a, Puryear & Gray 2024). Sharing experiences of suffering is a particularly promising intervention that connects with our harm-based moral minds. Even if we disagree about which acts are most harmful, we can all intuitively appreciate the fundamental moral instinct to protect the vulnerable from harm.

CONCLUSION

We began this review with a seeming moral paradox: Moral judgments vary across cultures and politics but seem to involve universal cognitive elements. We have suggested one way to resolve this paradox: by appreciating that we all share a common harm-based moral mind.

The harm of a harm-based mind is not the simple physical pain emphasized by early utilitarian accounts (Bentham 1789) or by some taxonomies of moral values (Haidt & Joseph 2007). Instead, harm is better understood as an intuitive perception constructed from the basic cognitive elements of perceived intention, causation of damage, and suffering (Schein & Gray 2018). This perception of interpersonal victimization lies at the core of human moral judgment.

Embracing our common harm-based moral mind does not require that we neglect cultural or political pluralism (Graham et al. 2013) nor does it lessen the value of studying differences in moral judgment between people. Moral diversity arises not in spite of universal concerns about harm but because of them: People from different cultures construct different perceptions of harm based on their unique beliefs and assumptions.

Appreciating our shared harm-based moral mind allows us to better understand moral disagreements across culture and politics. If you want to understand why your father-in-law votes differently from you, or why a foreign-exchange student emphasizes different values, your best bet is to understand their perceptions of harm.

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